

2-Way SSL Mutual Authentication AEM Forms JEE JBoss Turnkey Installation

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1. Create SSLMutualAuthProvider component in Eclipse or get pre-built copy from consulting team.

Creating Authentication Providers:

https://help.adobe.com/en_US/AEMForms/6.1/ProgramLC/WS624e3cba99b79e12e69a9941333732bac8-7fde.2.html

2. Deploy Component to AEM using Workbench

Using Workbench:

https://help.adobe.com/en_US/AEMForms/6.1/WorkbenchHelp/index.html

Managing Components & Services:

http://help.adobe.com/en_US/lifecycle/11.0/WorkbenchHelp/WS92d06802c76abadb7e4e0266128402897ed-7f77.2.html

3. Create Hybrid Domain in AdminUI. Domain name must match the name defined in the Authenticator SPI. Current default name is “cacDomain”.

Creating Domains:

http://help.adobe.com/en_US/lifecycle/11.0/AdminHelp/WS92d06802c76abadb-5145d5d12905ce07e7-7ffd.2.html

4. Create new group mapped to hybrid domain. Group name must match the name defined in the Authenticator SPI. Current default name is “cacGroup”

Creating Groups:

<https://helpx.adobe.com/experience-manager/6-4/forms/using/admin-help/creating-configuring-groups.html>

5. Create Keystore & TrustStore. Documentation for configuring 1-way SSL on JBoss located here:
https://help.adobe.com/en_US/AEMForms/6.1/AdminHelp/WS92d06802c76abadb-5145d5d12905ce07e7-7ff9.2.html
 - a. AEM 6.4:
 - i. <https://helpx.adobe.com/experience-manager/6-4/forms/using/admin-help/configuring-ssl-jboss-application-server.html>

- ii. <https://helpx.adobe.com/experience-manager/6-4/forms/using/admin-help/overview-6.html>

Server keystore and self-signed certificate may have been created during configuration manager process. If configured, cert and keystore is located (AEM 6): [Adobe Home]/jboss/standalone/configuration

Default keystore name is *aemformes.keystore* & default certificate name is *aemformes.cer*.

If SSL configuration was skipped you can run the configuration manager in [Adobe Home]/configurationManager/bin/ConfigurationManager.exe or create new keystore file and self-signed certificate as described below:

Create keystore file using keytool –genkey:

```
keytool -genkey -dname "CN=Host Name, OU=Group Name, O=Company Name, L=City Name, s=State, c=Country Code" -alias "LC Cert" -keyalg RSA -keypass key_password -keystore keystorename.keystore
```

Export the self-signed certificate file (lces.cer) using keytool –export

```
keytool -export -alias "LC Cert" -file lces.cer -keystore [appserver root]\server\[type]\conf\keystorename.keystore
```

For Mutual Authentication/2-Way SSL Implementation Use Steps 6-10:

6. Use or create a Trusted SSL Certificate Authority and corresponding personal certificate & private key (.p12) to use for client authentication. Testing certificates can be created using OpenSSL. Please contact the consulting team for assistance or to provide sample Trusted Root Certificate and corresponding client certs for testing.
7. Create client keystore/truststore file using same keytool –genkey command above. Alternatively, keystore can be created automatically during certificate import in next step.
8. Import Trusted SSL Root Certificate into client keystore/truststore file

- a. *keytool -import -trustcacerts -alias root -file .cer file -keystore client.keystore*

9. Copy client keystore/truststore to AEM JBoss configuration directory
10. Import Trusted SSL Root Certificate & .p12 file into your browser certificates.
 - a. Trusted Root Certificate should be imported first into *Trusted Root Certification Authorities*
 - b. Personal certificate (.p12) should be imported into *Personal*. After it's imported the name defined when creating the certificate should display in the "Issued To" column.
11. In order to map your local servers FQDN to localhost.
 - a. Add fqdn into Windows hosts file. Domain url can be anything but should be defined in the CN=[Host Name] field during keytool – genkey and in adminUI steps below.
 - b. Add URL information into C:\Windows\System32\drivers\etc\hosts
12. Add FQDN into Configure Referrer Whitelist page in AdminUI -- Settings/User Management/Configuration/Configure Allowed Referrer URL's
13. DRM Only: Update Base URL in Rights Management Server Configuration to [https://\[FQDN\]:8443](https://[FQDN]:8443). Uncheck all but last 2 checkboxes (Allow Client Certificate Authentication & Allow Extended Authentication). Update Extended Authentication Landing URL to [https://\[FQDN\]:8443](https://[FQDN]:8443)
14. Update JBoss XML configuration file to enable SSL Configuration. *Note: If SSL configuration was setup during Configuration Manager the SSL configuration will already include the self-signed server certificate information.*

ES4: Update *server.xml* located in:

jboss\server\lc_turnkey\deploy\jbossweb.sar. Uncomment the SSL/TSL connector configuration settings to enable 2-way SSL. Add the keystore/truststore file location paths along with their passwords.

Example:

```
<Connector protocol="HTTP/1.1" SSLEnabled="true" port="8443"
address="{jboss.bind.address}" scheme="https" secure="true" clientAuth="true"
keystoreFile="C:/Adobe/Adobe LiveCycle ES4/jboss/server/lc_turnkey/conf/lces.keystore"
```

```
keystorePass="password" truststoreFile="C:/Adobe/Adobe LiveCycle
ES4/jboss/server/lc_turnkey/conf/server.truststore"
truststorePass="password" sslProtocol = "TLS" />
```

AEM 6.1-6.3: Update *lc_turnkey.xml* located in [JBoss root]\standalone\configuration.

Example:

```
<connector name="http" protocol="HTTP/1.1" scheme="http" socket-binding="http"/>
<connector name="https" protocol="HTTP/1.1" scheme="https" socket-binding="https"
secure="true">
<ssl name="lc-ssl" password="testpass" verify-client="true" protocol="TLS" key-alias="LC Cert"
certificate-key-
file="C:/Adobe/Adobe_Experience_Manager_forms/jboss/standalone/configuration/lces.keystore"
ca-certificate-
file="C:/Adobe/Adobe_Experience_Manager_forms/jboss/standalone/configuration/client.keystore
" /></connector>
```

certificate-key-file: Full path to Server Keystore file

ca-certificate-file: Full path to Client Truststore file

password: server keystore password

AEM 6.4: Update *lc_turkey.xml* located in [JBoss root]\standalone\configuration

Update the following sections:

1. <Security-realm>
 - a. Add <authentication> for Mutual Authentication
2. <Https-listener> mapped to security realm defined in step 1.

Example:

keystore-path: Full path to Server Keystore file

keystore-password: keystore password

alias: certificate alias

<authentication>: Contains client truststore information

truststore path: Full path to Client Truststore file

keystore-password: truststore password

```
<security-realm name="SSLRealm">
<server-identities>
<ssl>
```

```
<keystore
path="C:/Adobe/Adobe_Experience_Manager_Forms/jboss/standalone/configuration/aemformses.keystore" keystore-password="password" alias="AEMformsCert" key-
password="password"/>
</ssl>
</server-identities>
<authentication>
<truststore
path="C:/Adobe/Adobe_Experience_Manager_Forms/jboss/standalone/configuration/clientTest.keystore" keystore-password="password"/>
</authentication>
</security-realm>
```

```
<https-listener name="default-secure" socket-binding="https" security-realm="SSLRealm"
verify-client="REQUIRED"/>
```

15. Restart JBoss Server

16. Try to access AdminUI using [https://\[fqdn\]:8443/adminui](https://[fqdn]:8443/adminui)